Statistical Physics for Machine Learning Exercise 3

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Abstract

The proposed exercise consists of comparing the curves of test error versus training dataset size for two perceptrons that take two 10-digits binary numbers (two 20-digits binary numbers for the second perceptron) and return 1 *if the first is bigger than the second,* -1 *otherwise. It is found that the ratio* P/N *is a universal parameter to describe the efficiency of the learning process. Moreover, we study the dependence of the test error on the noise parameter in the learning rule.*

Test error versus size of the training dataset

We build a perceptron that takes two 10-digits binary numbers via the mapping $0 \to -1$, $1 \to 1$ and returns 1 if the first is bigger than the second, -1 otherwise. Another perceptron is built with the same logic, but taking two 20-digits binary numbers. We use the randomised perceptron learning rule, that is we apply the following training strategy: If the predicted label is correct, we don't change the weights. If the predicted label is wrong, we change the weights according to:

$$\vec{J} = \vec{J} + \frac{1}{\sqrt{N}} \sigma_T^\mu \vec{\xi}^\mu (1 + \vec{\eta})$$

Where $\vec{\eta}$ is a random vector extracted from a N-dimensional gaussian distribution with mean $\vec{\mu} = \vec{0}$ and variance $\sigma^2 = 50$.

We train the first perceptron on datasets of size P = [1, 10, 20, 50, 100, 150, 200]. The second one is trained on datasets of size P = [1, 2, 20, 40, 100, 200, 300]. Then, we repeat the process 1000 times and we plot the test error ϵ versus the training dataset size P. We show the results in Figure (1). We note that the uncertainties in the test errors are too small to be visible.

We see that the $N = 40$ perceptron has a higher test error for any fixed training dataset size P. We can instead rescale the x-axis for the two perceptrons by considering P/N instead of P. Thus, we now study the test error ϵ as a function of the ratio between the training points and the number of parameters. The results are shown in Figure (2)

As we can see, the data now superimpose almost perfectly, well between the 2σ compatibility band highlighted in gray, although the agreement seems to get better and better as α grows larger and larger. These results suggest that the P/N ratio is a fundamental parameter that regulates how well the per-

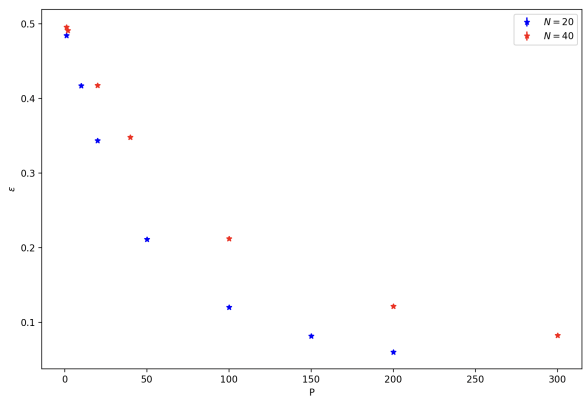

Figure 1: *Test error* ϵ *versus training dataset size* P *for perceptrons with* $N = 20$ *and* $N = 40$

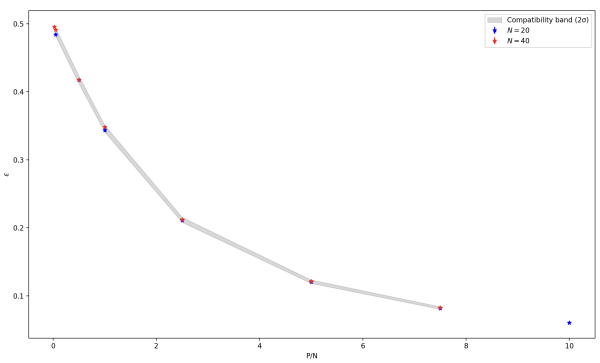

Figure 2: *Test error* ϵ *as a function of the ratio* P/N *with* 2σ *compatibility band*

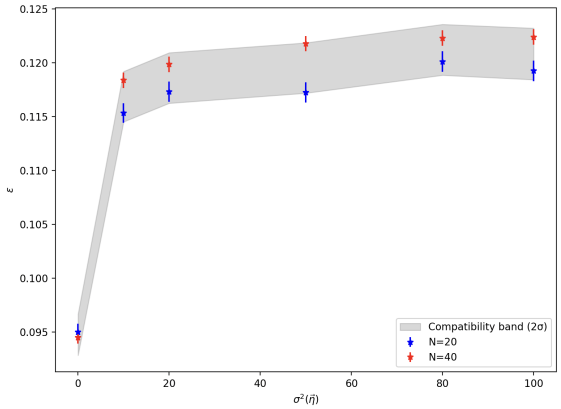

Figure 3: *Test error ϵ as a function of variance in the learning process*

ceptron learns the correct rule. We thus infer that the key element in assuring a low test error ϵ for our perceptron is to have a sufficiently high multiple $\alpha = P/N$ of training examples with respect to the number of parameters in our model.

Dependence of test accuracy on the noise in the training process

We now proceed to study how the noise in the training process, that is the variance of the random variable $\vec{\eta}$, affects the outcome of our perceptron: the test error ϵ. We fix the $\alpha = P/N$ ratio at $\alpha_0 = 5$ and we see how the test error changes when the variance of $\vec{\eta}$ takes the values $\sigma^2 = [0, 10, 20, 50, 80, 100]$. We do this both for the $N = 20$ and $N = 40$ perceptrons. The plots are shown in Figure (3)

As we can see, the test error seems to get quite worse as we go from the noiseless case to the noisy one, but the curve seems rather flat after $\sigma^2 = 20$. The noise parameter is typically used to avoid getting stuck in local minima, even though this time the learning process doesn't converge better (i.e. it doesn't converge to a 'better' minimum in the loss landscape) with increased noise. It might be more useful to implement a noisy learning rule when convergence is much harder due to a loss landscape that is significantly worse than the one in this particular example. Using a very high variance is however computationally demanding, in the sense that the convergence process is considerably slowed due to the big jumps that the student vector $\vec{J}$ makes in parameter-space, sometimes missing the local minima ($\vec{J}$s in version space) that would lead to a halt in the learning process (convergence).